EDS [1]
Wafer Test
/ Wafer Map

WAFER FLOW → Back Grinding → WAFER FLOW → Dicing → SINGULATED DIE FLOW → Die Attach → PACKAGE ASSEMBLY FLOW → Wire Bonding → PACKAGE ASSEMBLY FLOW → Molding / Marking

Wafer Map / Die Bin / Lot ID __ __ __ __ __ __ Good Die Selection by Wafer Map

PACKAGED DEVICE FLOW →

IMPLEMENTATION SCOPE [2]
PLC-BASED FINAL TEST HANDLER & MULTI-BIN SORTING CELL

Package Final Test

EDS electrical measurement and Wafer Map generation are simulated as upstream ATE data.

PLC controls material handling, interlock, test handshake, bin sorting, alarm handling, and production tracking.

PACKAGE / TRAY FLOW → Tray Packing / Shipment

[I/O DEVICE TAGS]

ES-01	Emergency Stop (NC)
PE-01	Infeed Presence Sensor
PE-02	Test Position Sensor
PE-03	Outfeed Confirm Sensor
MTR-01	Conveyor Motor
SOL-01	Test Clamp
SOL-02	PASS Gate
SOL-03	FAIL Gate
SOL-04	RETEST Gate
LT-01-G	Tower Lamp Green
LT-01-Y	Tower Lamp Yellow
LT-01-R	Tower Lamp Red
BZ-01	Buzzer

PACKAGE FINAL TEST & MULTI-BIN SORTING CELL

ST-01 INFEED
- Package Unit / Tray In
- Infeed Presence Sensor
- Conveyor Start Permission

→ ST-02 ALIGNMENT
- Presence Check
- Position Check
- Misalignment Interlock

→ ST-03 FINAL TEST NEST
- Stop Position
- Clamp / Fixture
- Test Request
- ATE Busy / Done
- Guard Door Interlock
- Clamp Closed Confirmation
- Product Present Confirmation
- Test Timeout Alarm

→ ST-04 MULTI-BIN SORTER
- Result Bin Validation
- Gate / Pusher Control
- PASS / FAIL / RETEST

PASS → BIN-1
FAIL → BIN-2
RETEST → BIN-3

→ ST-05 OUTFLOW
- Discharge Confirmation
- Bin Counter
- Tray Collection
- Gate Position Confirmation
- Outfeed Sensor Confirmation
- Jam Timeout Alarm

ATE-01 / MCU-01

ESP32-S3 Virtual ATE Simulator

- Test Result Simulation
- TEST_BUSY / TEST_DONE
- RESULT_BIN / ATE_FAULT
- HEARTBEAT

RS-485 / Modbus RTU

From PLC to ATE
TEST_REQ
TEST_ACK
RESET_REQ

From ATE to PLC
ATE_READY
TEST_BUSY
TEST_DONE
RESULT_BIN
ATE_FAULT
HEARTBEAT

PLC-01

- LS XBC-DR32H
- Main Cell Controller

- Sequence / Interlock
- Conveyor / Clamp / Gate
- Alarm / Safety / Counter

RS-232C

PLC Status
Alarm
Production Data
HMI Command Request

SCADA-01

- LabVIEW SCADA / HMI

- Mode / Status / Alarm
- Production Count / Log
- Cycle Time / Trend

LEGEND
- Material Flow
- Data Flow
- Control / Handshake
- Implementation Scope

[DESIGN NOTES]

[1] EDS is an upstream wafer-level test process. Wafer Map / Die Bin / Lot ID are simulated as upstream ATE data in this project.

[2] This project physically implements only the Package Final Test Handler and Multi-Bin Sorting Cell.

[3] PLC has final authority over sequence, safety interlock, alarm, and all actuator outputs. LabVIEW is used for SCADA/HMI and logging.

BIN DEFINITION

Bin	Result	Destination	PLC Action
1	PASS	BIN-01	PASS Gate Open
2	FAIL	BIN-02	FAIL Gate Open
3	RETEST	BIN-03	RETEST Gate Open
9	INVALID	Quarantine / Stop	Alarm and Hold

NO.	DATE	REVISION
A	2026-08-04	INITIAL RELEASE

DWG TITLE
EDS Wafer Map Linked Package Final Test and Multi-Bin Sorting Cell
Process Flow and System Architecture

DRAWN BY: J. H. KO
DATE: 2026-08-04
DWG NO: 002
SHEET NO: 1 OF 1